

Exp No 2

Study/Demo of 5 axis robotic arm

Aim: Study/Demo of 5 axis robotic arm

A 5-axis robot is a robotic arm that can move in five directions: up and down, left and right, forward and backward, yaw, and pitch. It's considered the pinnacle of robotic engineering because of its precision and adaptability.

How does a 5-axis robot work?

- Each axis is controlled by a motor.
- The axes are represented by X, Y, Z, A, and B.
- The robot's articulated joints allow for a wide range of movements.
- The robot's flexibility and dexterity enable tasks that were once too complex for traditional robotic systems.
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What are the uses of a 5-axis robot?

- 5-axis robots can create more complex shapes.
- They can improve tolerances.
- They can reduce the need for time-consuming changeovers.
- They can be used for pick and place operations.

Result:

